

A009 · Integrating pharmacogenomics and cheminformatics with diverse disease phenotypes for cell type-guided drug discovery

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Background: Large-scale pharmacogenomic resources, such as the Connectivity Map (CMap), have greatly assisted computational drug discovery. However, despite their widespread use, CMap-based methods have thus far been agnostic to the biological activity of drugs as well as to the genomic effects of drugs in multiple disease contexts. Here, we present a network-based statistical approach, Pathopticon, that uses CMap to build cell type-specific gene-drug perturbation networks and integrates these networks with cheminformatic data and diverse disease phenotypes to prioritize drugs in a cell type-dependent manner.

Methods: We build cell type-specific gene-drug perturbation networks from CMap data using a statistical procedure we call Quantile-based Instance Z-score Consensus (QUIZ-C). Using these networks and a large-scale disease-gene network consisting of 569 disease signatures from the Enrichr database, we calculate Pathophenotypic Congruity Scores (PACOS) between input gene signatures and drug perturbation signatures and combine these scores with cheminformatic data from ChEMBL to prioritize drugs. We benchmark our approach by calculating area under the receiver operating characteristic curves (AUROC) for 73 gene sets from the Molecular Signatures Database (MSigDB) using target gene expression profiles from the Comparative Toxicogenomics Database (CTD). We validate the drugs predicted in our proofs-of-concept using real-time polymerase chain reaction (qPCR) experiments.

Results: Cell type-specific gene-drug perturbation networks built using QUIZ-C are topologically distinct, reflecting the biological uniqueness of the cell lines in CMap, and are enriched in known drug targets. Pathopticon demonstrates a better prediction performance than solely cheminformatic measures as well as state-of-the-art network and deep learning-based methods. Top predictions made by Pathopticon have high chemical structural diversity, suggesting their potential for building compound libraries. In proof-of-concept applications on vascular diseases, we demonstrate that Pathopticon helps guide in vitro experiments by identifying pathways that are potentially regulated by the predicted therapeutic candidates.

Conclusions: Our network-based analytical framework integrating pharmacogenomics and cheminformatics (available at <https://github.com/r-duh/Pathopticon>) provides a feasible blueprint for a cell type-specific drug discovery

and repositioning platform with broad implications for the efficiency and success of drug development.

Integrating pharmacogenomics and cheminformatics with diverse disease phenotypes for cell type-guided drug discovery

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